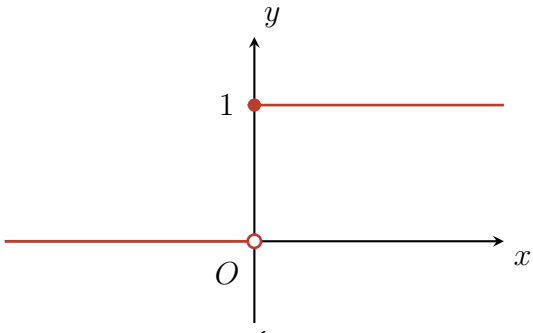
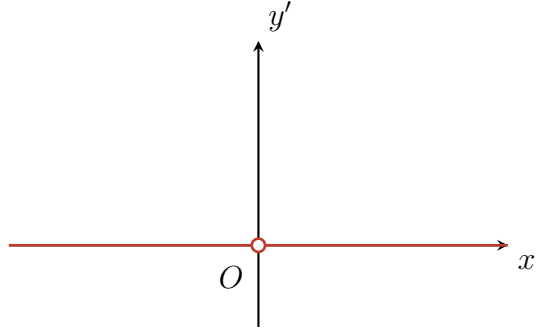


$$f(x)=\begin{cases}0, & x<0\\ 1, & x\geq 0\end{cases}$$



$$f'(x)=0 \quad (x\neq 0)$$